

6. Infrastructure Requirements & Sandbox Deployment Specifications

6.1. Network Topology Tier (The PNETLab VM)

Because PNETLab runs virtual network nodes as nested virtual machines, this single VM will act as the heavy lifter for the 20 virtual switches.

Per-Switch Footprint (Dell OS10):

A single virtual Dell OS10 instance typically requires 1 to 2 vCPUs and 3 GB to 4 GB of RAM to boot comfortably and run control plane functions smoothly (like LLDP, RESTCONF, and gNMI).

Total Topology Requirements:

- vCPUs: 30 to 40 vCPUs (Assuming an overcommit ratio of roughly 1.5:1 to 2:1 since switches aren't handling real multi-gigabit traffic).
- RAM: 64 GB to 80 GB dedicated strictly to the switches.
- Storage: 120 GB to 150 GB of fast NVMe/SSD space (each virtual image takes roughly 5-6 GB of disk allocation).

Mandatory Infrastructure Prerequisite: Activation of Nested Virtualization (VT-x/AMD-V) is required at the hypervisor level (ESXi, Proxmox, or GCP). Failure to enable this feature forces QEMU into software emulation mode, preventing the Dell OS10 virtual instances from successfully completing their boot sequence.

6.2. Application & Cluster Tier (The 3 VMs + Load Balancer)

Based on the platform specs, this plane includes a FastAPI Gateway, 4-Stage Policy Engines, Celery Workers, a PostgreSQL Database, and a Redis Sentinel HA Cluster.

Distributing this load across 3 Application VMs managed by 1 Load Balancer (LB) requires the following allocation:

VM 1, 2, & 3: Application & Database Nodes (Active-Active-Active + Sentinel)

- vCPU: 4 vCPUs per VM (12 vCPUs total). Needed to manage background asyncio loops for LLDP discovery, metrics parsing, and concurrent Celery task executions.
- RAM: 8 GB of RAM per VM (24 GB total). This gives breathing room for running Redis Sentinel, local FastAPI workers, and the primary/replica PostgreSQL instances.
- Storage: 40 GB SSD per VM. This accommodates the underlying OS and logs. (The shared PostgreSQL tables-like topology_edges and discovered_endpoints-are highly transactional but physically compact).

VM 4: Load Balancer (HAProxy, NGINX, or a lightweight LB VM)

- vCPU: 2 vCPUs
- RAM: 2 GB to 4 GB of RAM
- Storage: 20 GB SSD

6.3. Resource Requirements & Infrastructure Matrix

For a seamless deployment of the full SDN ecosystem within a sandbox environment, the host infrastructure should meet the following aggregate resource specifications:

SDN Ecosystem Deployment Specifications

Architectural Component	Unit	vCPU Profile	RAM Quota	Storage Cap
PNETLab Topology VM (20 Switches)	1	32-40 vCPUs	64 GB - 80 GB	150 GB NVMe
SDN Application Plane (FastAPI/Celery)	3	4 vCPUs (12 Tot)	8 GB (24 GB Tot)	40 GB SSD / node
Core Load Balancer VM	1	2 vCPUs	4 GB	20 GB SSD
TOTAL SOLUTION ESTIMATE	5 VMs	~46 to 54 vCPUs	~92 to 108 GB	~300 GB SSD